CHP 1: MAGNETOSTATIC FIELDS:

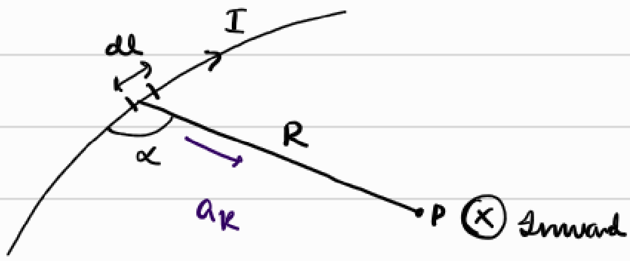
→ Electrostatic field by static charges, magnetostatic field produced by charges moving with constant velocity → produced by constant current flow (direct current)

→ Biot Savart's law → Columb's law $\qquad H \rightarrow E \qquad$ $B = \mu_0 H$

Ampere Circuital law → Gauss law $\qquad B \rightarrow D \qquad$ $D = \varepsilon_0 E$

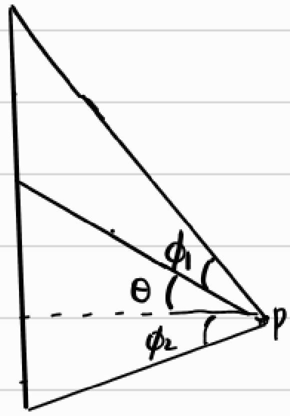
→ Biot Savart's law:

$$\underbrace{dH}_{\text{magnetic field intensity}} \propto \frac{\overbrace{I\,dl}^{\text{differential current element}} \sin \underbrace{\alpha}_{\text{angle b/w the element and the line}}}{\underbrace{R^2}_{\text{distance b/w P and the element}}}$$



$$dH = \frac{K I\,dl \sin\alpha}{R^2} \Rightarrow \frac{1}{4\pi} \frac{I\,dl \sin\alpha}{R^2} \qquad \Rightarrow K = \frac{1}{4\pi}$$

$$dH = \frac{1}{4\pi} \frac{I\,dl \times a_r}{R^2} \Rightarrow \frac{1}{4\pi} \frac{I\,dl \times \vec{R}}{R^3}$$



→ ⊙ → Out of the page

⊗ → into the page

→ different current distributions:

- ↳ line current $\qquad H = \int \frac{I\,dl \times a_r}{4\pi R^2} \qquad$ I → line current density
- ↳ surface current $\qquad H = \int \frac{K\,ds \times a_r}{4\pi R^2} \qquad$ K → surface current density
- ↳ volume current $\qquad H = \int \frac{J \cdot dv \times a_r}{4\pi R^2} \qquad$ J → volume current density

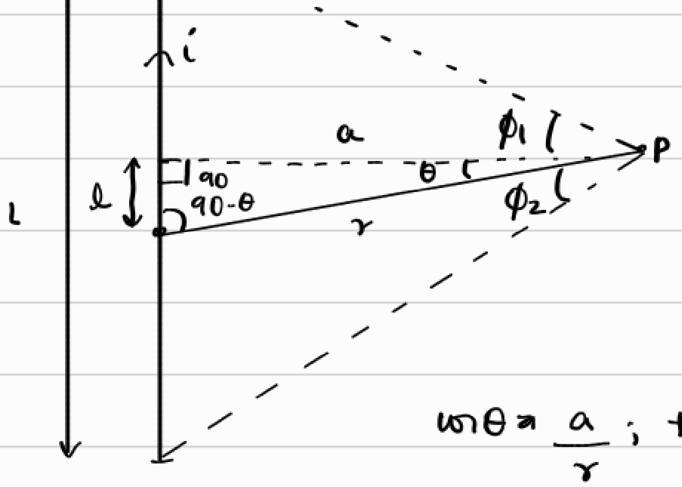
$$I \cdot dl = K \cdot ds = J \cdot dV$$

→ $\vec{dB} = \mu_0 \cdot \vec{dH} = \frac{\mu_0}{4\pi} \frac{I\,dl \times \vec{R}}{R^3} \longrightarrow \frac{\mu_0}{4\pi} = 10^{-7}$; $\mu = \mu_r \cdot \mu_0$

→ which configuration you take, the cross product matters

→ Magnetic field due to finite straight current carrying wire:

$dB = \mu_0 \quad I \cdot dl \times R$



$\frac{\quad}{4\pi}\ \frac{\quad}{R^3}$

$\Rightarrow \frac{\mu_0}{4\pi}\ \frac{I\,dl\,\sin(90-\theta)}{R^2}$

$\Rightarrow \frac{\mu_0}{4\pi R^2}\cdot I\cdot dl\cdot\cos\theta$

$\cos\theta = \frac{a}{r}$; $\tan\theta = \frac{l}{a} \Rightarrow l = a\tan\theta \Rightarrow dl = a\sec^2\theta\, d\theta$

$\vec{dB} = \frac{\mu_0}{4\pi}\cdot\frac{I\cdot a\,\cancel{\sec^2\theta}\cdot\cos\theta\cdot d\theta}{(a/\cancel{\cos\theta})^2} = \int_{-\phi_2}^{\phi_1}\frac{\mu_0}{4\pi}\cdot\frac{I}{a}\cos\theta\cdot d\theta$

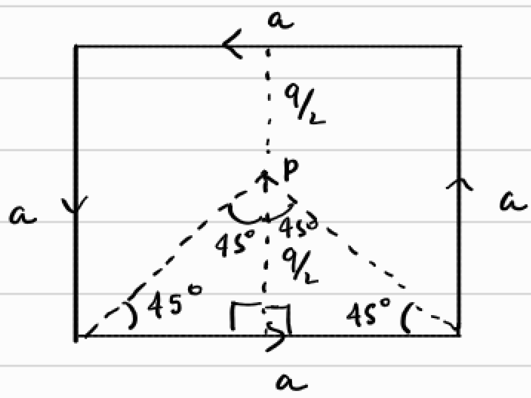
$\Rightarrow \frac{\mu_0}{4\pi}\cdot\frac{I}{a}\int_{-\phi_2}^{\phi_1}\cos\theta\cdot d\theta$

$\Rightarrow \frac{\mu_0}{4\pi}\cdot\frac{I}{a}\left[\sin\theta\right]_{-\phi_2}^{\phi_1}$

$= \frac{\mu_0}{4\pi}\frac{I}{a}\left[\sin\phi_1 - \sin(-\phi_2)\right] \Rightarrow \frac{\mu_0}{4\pi}\cdot\frac{I}{a}\left[\sin\phi_1 + \sin\phi_2\right]$

$\therefore \vec{B} = \frac{\mu_0}{4\pi}\cdot\frac{I}{a}\left[\sin\phi_1 + \sin\phi_2\right] a_a$

(a)



$\Rightarrow B_1 = B_2 = B_3 = B_4$

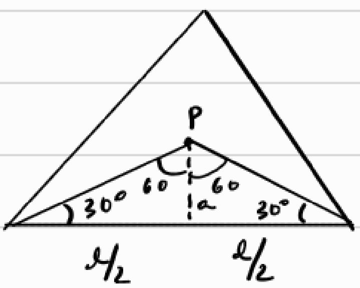
$B_T = 4B_1$

$B_1 \Rightarrow \frac{\mu_0}{4\pi}\cdot\frac{I}{a}\left[\sin\phi_1 + \sin\phi_2\right]$

$\Rightarrow \frac{\mu_0}{4\pi}\cdot\frac{I}{a/2}\left[\sin(\pi/4) + \sin(\pi/4)\right]$

$\Rightarrow \frac{\mu_0}{4\pi}\cdot\frac{2\sqrt{2}I}{a} \Rightarrow B_1 \Rightarrow B_T = \cancel{4}\times\frac{\mu_0}{\cancel{4}\pi}\cdot\frac{2\sqrt{2}I}{a} = \frac{\mu_0}{\pi}\cdot\frac{2\sqrt{2}I}{a}\, a_a$

(a)



$B_T = 3B_1$

$\Rightarrow \tan 30° = \frac{a}{l/2} \Rightarrow a \Rightarrow \frac{l}{2}\cdot\frac{1}{\sqrt{3}} = \frac{l}{2\sqrt{3}}$

$B_1 \Rightarrow \frac{\mu_0}{4\pi}\cdot\frac{I}{a}\left[\sin\phi_1 + \sin\phi_2\right]$

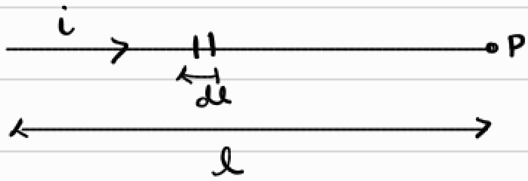
$$B_1 = \frac{\mu_0}{4\pi} \cdot \frac{I \cdot 2\sqrt{3}}{l}\left[\sin 60^\circ + \sin 60^\circ\right] = \frac{\mu_0}{4\pi} \cdot \frac{I \cdot 2\sqrt{3} \cdot \sqrt{3}}{l} = 6 \cdot \frac{\mu_0}{4\pi} \frac{I}{l}$$

$$B_T = 3B_1 = \frac{9}{2} \frac{\mu_0}{\pi} \frac{i}{l} \, a_a$$

→ Magnetic field due to finite straight current carrying wire

4 CASES:

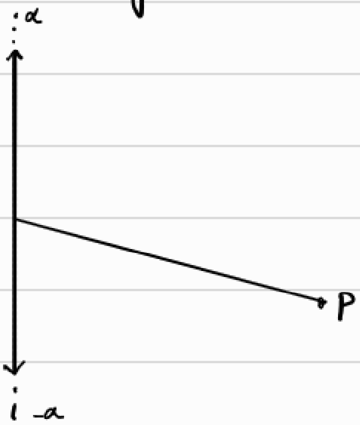
(1)



$$\vec{B}' = \frac{\mu_0}{4\pi} \cdot \frac{I}{a}\left[\sin 0 + \sin 0\right]$$

$$\vec{B}' = 0$$

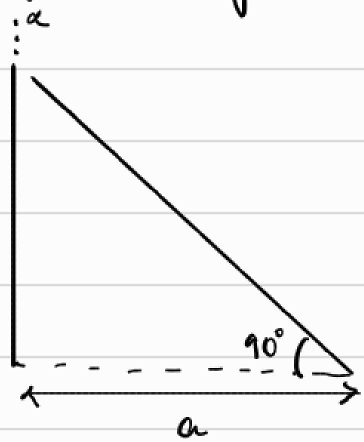
(2) infinite straight current carrying wire



$$\vec{B}' = \frac{\mu_0}{4\pi} \cdot \frac{I}{a}\left[\sin \pi/2 + \sin \pi/2\right]$$

$$= \frac{\mu_0}{2\pi} \cdot \frac{I}{a} \, a_a$$

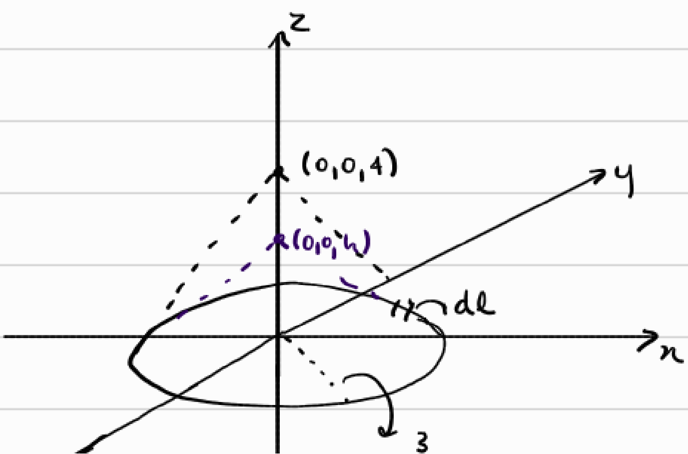
(3) semi infinite straight current carrying wire



$$\vec{B}' = \frac{\mu_0}{4\pi} \cdot \frac{I}{a}\left[\sin \pi/2 + \sin 0\right]$$

$$= \frac{\mu_0}{4\pi} \cdot \frac{I}{a} \, a_a$$

(Q) Circular loop: $x^2+y^2=9$; $z=0$, $I = 10 a_\phi$



$$dH = \frac{I \, dl \times R}{4\pi R^3}$$

$dl = \rho \cdot d\phi \, a_\phi$

$R = (0,0,h) - (x,y,0)$

$= h a_z - \rho a_\rho$

$dl \times R = |$

$$dl \times R = \begin{vmatrix} a_\rho & a_\phi & a_z \\ 0 & \rho d\phi & 0 \\ -\rho & 0 & h \end{vmatrix}$$

$$= \rho h d\phi\, a_\rho + \rho^2 d\phi\, a_z$$

$$dH = \frac{I}{4\pi[\rho^2+h^2]^{3/2}} (\rho h d\phi\, a_\rho + \rho^2 d\phi\, a_z)$$

$$= \cancel{dH_\rho} + dH_z$$

$dH_\rho \to 0.$ all radial components will cancel out

$$H = \int dH_z\, a_z = \int_0^{2\pi} \frac{I}{4\pi[\rho^2+h^2]^{3/2}} \cdot \rho^2 \cdot d\phi \cdot a_z = \frac{I\rho^2}{\cancel{4\pi}_{\,2}\,[\rho^2+h^2]^{3/2}} \cdot \cancel{2\pi}$$

$$H = \frac{I\rho^2}{2[\rho^2+h^2]^{3/2}}\, a_z$$

(a) $I = 10A$, $h = 4$

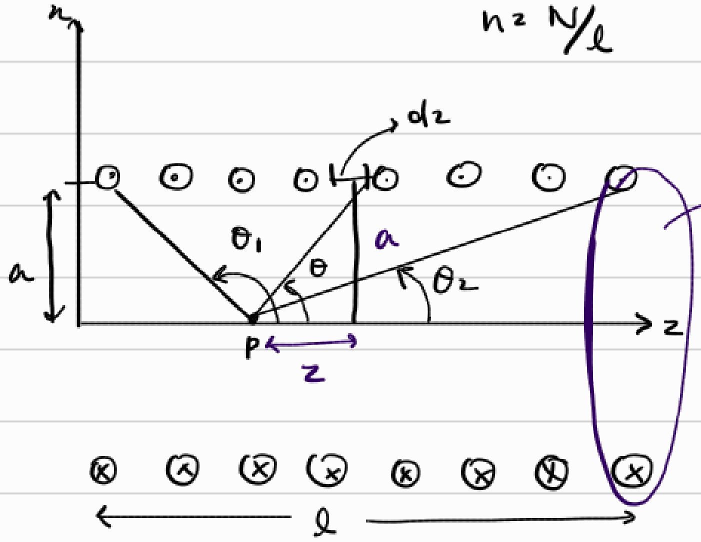
$$H = 0.36\, a_z \text{ A/m}$$

(b) $I = 10A$, $h = -4$

$$H = 0.36\, a_z \text{ A/m}$$

(Q) Solenoid: → length l, radius a, Total turn N, current I

$$n = N/l$$



circular loop (many are present)

∴ Total field = ∫(field from each loop)

$$dH_z = \frac{I \cdot dl\, a^2}{2[a^2+z^2]^{3/2}} = \frac{I \cdot a^2 n dz}{2[a^2+z^2]^{3/2}}$$

$dl = n \cdot dz$ where $n = N/l$.

dl → small thickness

$$\tan\theta = \frac{a}{\ } \Rightarrow z = a\cot\theta$$

z $\quad dz = -a\,\mathrm{cosec}^2\theta\, d\theta$

$$\sin\theta = \frac{a}{(a^2+z^2)^{1/2}} \Rightarrow \sin^3\theta = \frac{a^3}{(a^2+z^2)^{3/2}} \Rightarrow (a^2+z^2)^{3/2} = \frac{a^3}{\sin^3\theta}$$

$$dH_z = \frac{I\sin^3\theta \cdot \cancel{a^2} \cdot n \cdot -\cancel{a}\,\mathrm{cosec}^2\theta \cdot d\theta}{2\cancel{a^3}} = -\frac{In}{2}\sin\theta \cdot d\theta$$

$$H_z = -\frac{nI}{2}\int_{\theta_1}^{\theta_2} \sin\theta \cdot d\theta = \frac{nI}{2}\left[-\cos\theta_1 + \cos\theta_2\right]$$

$$= \frac{nI}{2}\left[\cos\theta_2 - \cos\theta_1\right] a_z$$

Field at the center of the solenoid:

$\cos\theta_1 = -\cos\theta_2$

$$H = \frac{nI}{2}\left[\cos\theta_2 + \cos\theta_2\right] = \frac{nI}{2} \cdot 2\cos\theta_2 = nI\cos\theta_2 = nI \cdot \frac{l/2}{\sqrt{a^2 + l^2/4}}\, a_z$$

$\longrightarrow l >>> a \quad \theta = 0°$ and $180°$

$$\therefore \quad H = \frac{nI}{2}\left[\cos 0 - \cos 180\right] = nI\, a_z = \frac{NI}{l} a_z$$

Ampere's Circuital law – Maxwell's Equation:

$$\oint_L H \cdot dl = I_{enc}$$

↳ used to determine H only when current distribution is symmetrical

Apply Stokes theorem:

$$\underbrace{\oint_L H \cdot dl = \int_S (\nabla \times H) \cdot ds}_{\text{Ampere's law in integral form}} \Rightarrow I_{enc}.$$

$$I_{enc} = \int_S J \cdot ds$$

$$\therefore \int_S (\nabla \times H) \cdot ds = \int_S J \cdot ds \Rightarrow \nabla \times H = J \longrightarrow 3^{rd} \text{ maxwell equation}$$

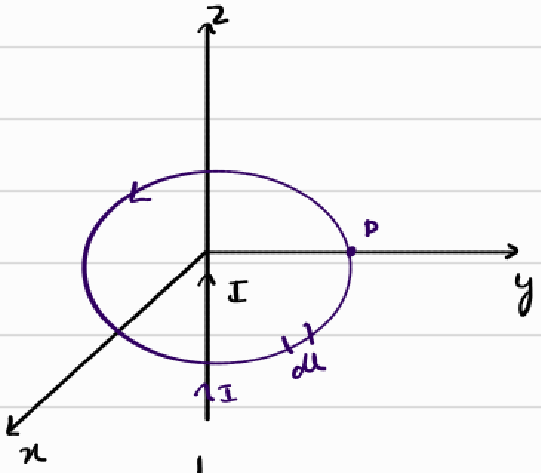
↳ Ampere's law in differential (point) form magnetostatic field is not conservative

Applications of Ampere's law:

$\otimes \rightarrow +ve$ $\quad \longrightarrow \oint_c B\cdot dl = \mu_0 \cdot i_{inside}$

$\odot \rightarrow -ve$

(A) Infinite line current:

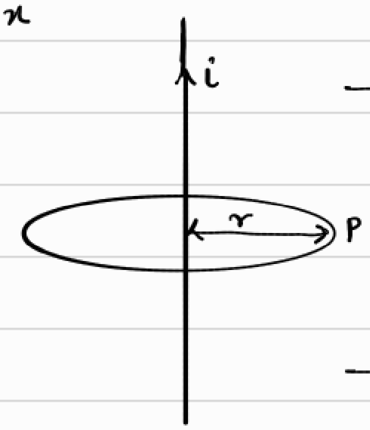


$\oint_L H\cdot dl = I_{enc}$ $\quad H = H_\phi a_\phi$

$dl = \rho d\phi a_\phi$

$= \int_L H_\phi a_\phi \cdot \rho \cdot d\phi a_\phi = H_\phi \int_L \rho \cdot d\phi = H_\phi \cdot 2\pi\rho$

$H = \frac{I}{2\pi\rho}$



$\longrightarrow B = \frac{\mu_0}{4\pi}\cdot\frac{I}{r}\,[\sin 90 + \sin 90]$

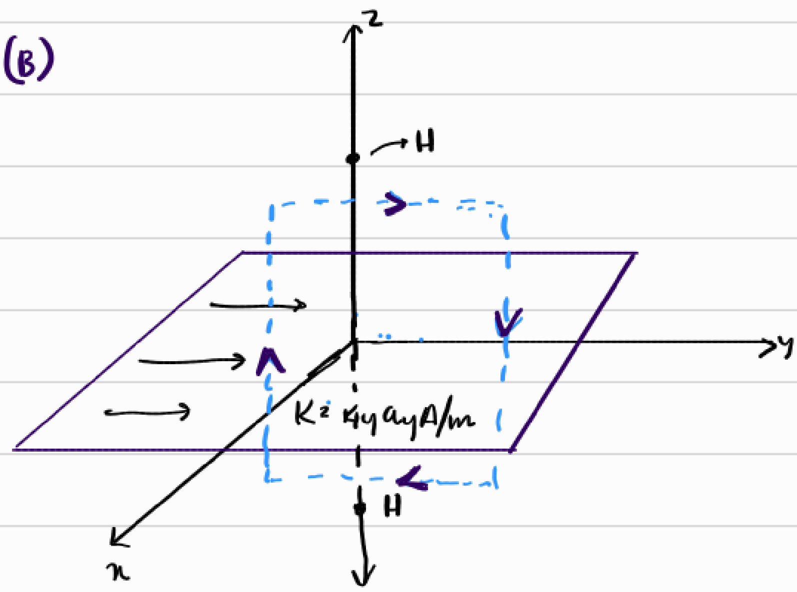
$= \frac{\mu_0}{\cancel{4}_2\pi}\cdot\frac{\cancel{2}I}{r} = \frac{\mu_0 I}{2\pi r} a_r$

$\longrightarrow \oint B\cdot dl = \mu_0 \cdot i_{inside}$

$B\int dl = \mu_0 \cdot i_{inside}$

$B\cdot 2\pi r = \mu_0 \cdot i_{inside} \Rightarrow B = \frac{\mu_0 I}{2\pi r}\hat{a_r}$

(B)



$\oint H\cdot dl = I_{enc} = \int K\cdot dl$

rectangular loop crossing the sheet [half above and half below]

Top side: Field and path are parallel $(H\cdot L)$

Bottom side: Field and path are antiparallel $[-H\cdot -L]$

$= [H\cdot L]$

Vertical sides: 0.

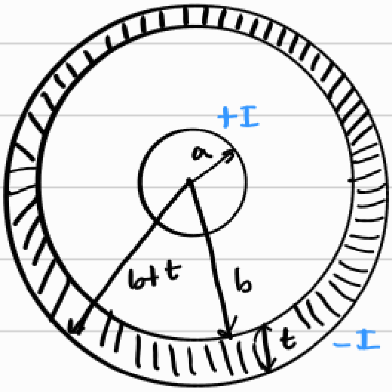
$\therefore \oint H\cdot dl = 2HL = I_{enc} = K\cdot L$

$\therefore H = \frac{K}{2} \Rightarrow H = \pm \frac{K}{2} a_n$ → depending on the direction of the current charge density

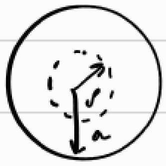
$B = \mu_0 \cdot H = \frac{\mu_0 \cdot K}{2}$

$H = \frac{1}{2} K \times a_n$ where a_n → normal unit vector from the current sheet to the point of interest

(c) Infinitely long coaxial transmission line:



(i) $\rho \leq a$



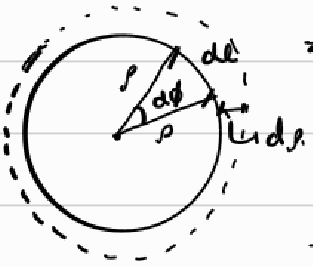
$J = \frac{I}{\pi a^2}$ (volume density)

$ds = \rho \cdot d\rho \cdot d\phi$

$\oint H \cdot dl = I_{inside} = \int J \cdot ds$

$\Rightarrow H \int_0^{2\pi} \rho \cdot d\phi = \int_0^{2\pi} \int_0^{\rho} \frac{I}{\pi a^2} \rho \cdot d\rho \cdot d\phi$

$J = \frac{dI}{dA}$

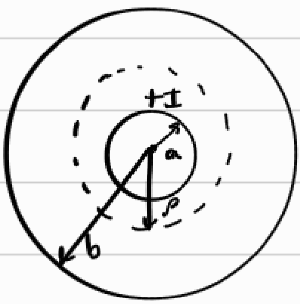


$\Rightarrow \rho \cdot d\phi = dl \times d\rho \Rightarrow \rho \cdot d\phi \cdot d\rho$

$\Rightarrow H \int_0^{2\pi} \rho \cdot d\phi = \frac{I}{\cancel{\pi} a^2} \cdot \frac{\rho^2 \cancel{2\pi}}{\cancel{2}}$

$\Rightarrow H \cdot 2\pi \cancel{\rho} = \frac{I}{a^2} \cdot \rho^{\cancel{2}} \Rightarrow H = \frac{I \cdot \rho}{2\pi a^2} a_\phi$

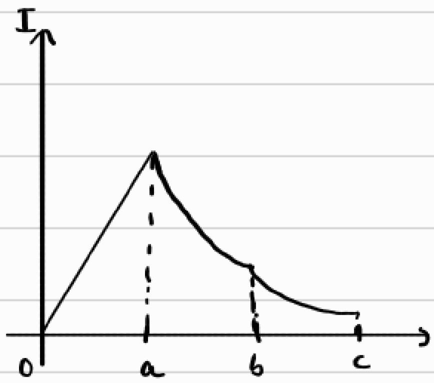
(ii) $a \leq \rho < b$



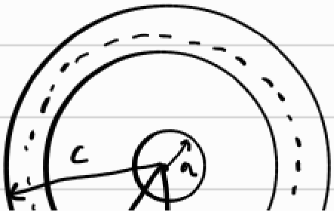
$\oint H \cdot dl = I_{enclosed}$

$H \cdot 2\pi\rho = I$

$\vec{H} = \frac{I}{2\pi\rho} a_\phi$



(iii) $b \leq \rho \leq b + t = c$



$\oint H \cdot dl = I_{enc} = +I - I\left[\frac{\pi\rho^2 - \pi b^2}{\pi c^2 - \pi b^2}\right]$

$J = \frac{I}{\pi c^2 - \pi b^2}$

$ds = \pi\rho^2 - \pi b^2$

$H \int_0^{2\pi} \rho \cdot d\phi = I\left[1 - \left[\frac{\cancel{\pi}\rho^2 - \cancel{\pi}b^2}{\ldots}\right]\right]$

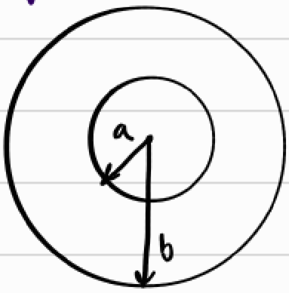
$$H \cdot 2\pi\rho = I\left[\frac{c^2-\rho^2}{c^2-b^2}\right]$$

$$\vec{H} = \frac{I}{2\pi\rho}\left[\frac{c^2-\rho^2}{c^2-b^2}\right]\hat{a}_\phi$$

(iv) $\rho > b$

$$-I + I = 0 \Rightarrow \vec{H} = 0$$

(D) Hollow Conducting cylinder:



(i) $\rho < a$

$$\oint H \cdot dl = I = 0$$

$$\therefore \vec{H} = 0$$

(ii) $a < \rho < b$

$$J = \frac{I}{\pi b^2 - \pi a^2} \Rightarrow I = \int_a^\rho J \cdot ds = \frac{I\cancel{\pi}(\rho^2-a^2)}{\cancel{\pi}b^2 - \cancel{\pi}a^2}$$

$$H \cdot 2\pi\rho = \frac{I(\rho^2-a^2)}{(b^2-a^2)} \Rightarrow \vec{H} = \frac{I}{2\pi\rho}\left[\frac{\rho^2-a^2}{b^2-a^2}\right]\hat{a}_\rho$$

→ Magnetic Scalar and Vector potentials:

$$\nabla V = -E$$

magnetic potential associated with magnetostatic field B can be scalar V_m or vector A

$$\left.\begin{array}{l}\nabla \times (\nabla V) = 0 \\ \nabla \cdot (\nabla \times A) = 0\end{array}\right] \rightarrow \text{must hold for any scalar field and vector field}$$

→ $H = -\nabla V_m$ if $J = 0$ → $V_m = -\int_A^B \vec{H} \cdot \vec{dl}$ [Ampere's law not valid ∵ $J=0$]

→ $J = \nabla \times H \Rightarrow \nabla \times (-\nabla V_m) = 0$ → magnetic scalar potential only defined in the region $J=0$

→ $\nabla^2 V_m = 0$ $(J=0)$ → scalar laplacian equation

→ $\nabla \cdot B = 0$ (for magnetostatic field)

$\longrightarrow \quad B = \nabla \times A \longrightarrow$ magnetic vector potential A

$$V = \int \frac{dq}{4\pi\varepsilon_0 R} \longrightarrow \text{similarly:} \quad A = \int_L \frac{\mu_0 I dl}{4\pi R} \longrightarrow \text{line current}$$

$$A = \int_S \frac{\mu_0 K \cdot ds}{4\pi R} \longrightarrow \text{surface current}$$

$$A = \int_V \frac{\mu_0 J \cdot dv}{4\pi R} \longrightarrow \text{volume current}$$

Obtaining eq^n of line current, surface current and volume current from:

$$B = \frac{\mu_0}{4\pi} \int \frac{I \cdot dl' \times R}{R^3}$$

where R is the distance vector from line element dl' (x', y', z') to the point $R(x, y, z)$

$$R = r - r' = [(x-x')^2 + (y-y')^2 + (z-z')^2]^{1/2}$$

$$\nabla\left(\frac{1}{R}\right) = -\frac{\vec{R}}{R^3}$$

$$B = \frac{\mu_0}{4\pi} \int I \cdot dl' \times \nabla\left(\frac{1}{R}\right) \qquad \nabla \times (fF) = f\,\nabla \times F + (\nabla f) \times F$$

$$f = \frac{1}{R} \quad F = dl'$$

$$dl' \times \nabla\left(\frac{1}{R}\right) = -\nabla \times \left(\frac{dl'}{R}\right)$$

$$B = \nabla \times \underbrace{\int_L \frac{\mu_0}{4\pi} \cdot \frac{I \cdot dl'}{R}}_{A}$$

Magnetic flux: Ψ

$$\hookrightarrow \Psi = \int_S B \cdot ds = \int_S (\nabla \times A) \cdot ds = \oint_L A \cdot dl$$

$$\Psi = \oint_L A \cdot dl$$

$$\vec{B} = \nabla \times A$$

$$\nabla \times H = \vec{J} \qquad \nabla \times \vec{B} = \mu_0 \cdot \vec{J}$$

$$\nabla \times (\nabla \times A) = \cancel{\nabla (\nabla \cdot \vec{A})} - \nabla \vec{A} = \boxed{\nabla^2 \vec{A} \neq 0}$$

$$\therefore \nabla \times \vec{B} = \nabla^2 \vec{A} = -\mu_0 J$$

$$\boxed{\nabla^2 A = -\mu_0 J} \longrightarrow$$ vector poisson equation in magnetostatics

$$\boxed{\nabla^2 V_m = 0} \longrightarrow$$ laplacian equation

$\longrightarrow$ Derivation of Biot-savart's law and Ampere's law:

Biot savart law:

$$\nabla \times (\nabla \times A) = \nabla \cdot (\nabla \cdot A) - \nabla^2 A$$

$$B = \nabla \times \oint \frac{\mu_0 \, dl'}{4\pi R} = \frac{\mu_0 I}{4\pi} \oint \nabla \times \left(\frac{dl'}{R} \right) \quad \begin{cases} F = dl' \\ f = 1/R \end{cases}$$

$$= \frac{\mu_0 I}{4\pi} \oint \left[\frac{1}{R} \cancel{\nabla \times dl'} + \left(\nabla \tfrac{1}{R} \right) \times dl' \right]$$

$\nabla \rightarrow x, y, z$

$dl' \rightarrow x', y', z'$

$$1/R = \left[(x-x')^2 + (y-y')^2 + (z-z')^2 \right]^{-1/2}$$

$$\nabla \left(\frac{1}{R} \right) = \frac{(x-x')a_x + (y-y')a_y + (z-z')a_z}{\left| (x-x')^2 + (y-y')^2 + (z-z')^2 \right|^{3/2}} = \frac{-a_R}{R^2}$$

$$B = \frac{\mu_0 I}{4\pi} \oint \frac{dl' \times a_R}{R^2}$$

$$\nabla \times B = \nabla \times (\nabla \times A) = \cancel{\nabla \cdot (\nabla \cdot A)}^{\,0} - \nabla^2 A \rightarrow \nabla \cdot A \rightarrow \text{Coloumb's gauge}$$

$$\nabla \times B = -\nabla^2 A = \mu_0 J$$

$$\left. \begin{aligned} \nabla^2 A_x &= -\mu_0 J_x \\ \nabla^2 A_y &= -\mu_0 J_y \\ \nabla^2 A_z &= -\mu_0 J_z \end{aligned} \right] \rightarrow$$ scalar Poisson equation

Ampere Circuital law:

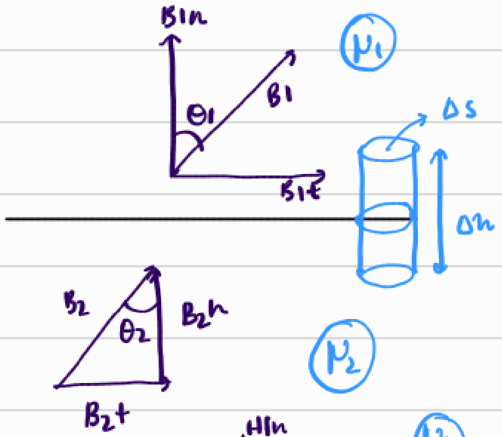
$$\oint H \cdot dl = \int_s (\nabla \times H) \cdot ds = \frac{1}{\mu_0} \int \nabla \times (\nabla \times A) \cdot ds = \frac{1}{\cancel{\mu_0}} \cdot -\cancel{\mu_0} \int J \cdot ds$$

$\therefore \oint H \cdot dl = \int J \cdot ds = I_{enc}$

→ **Magnetic Boundary condition:**

$\oint_s B \cdot ds = 0$ → for a closed surface

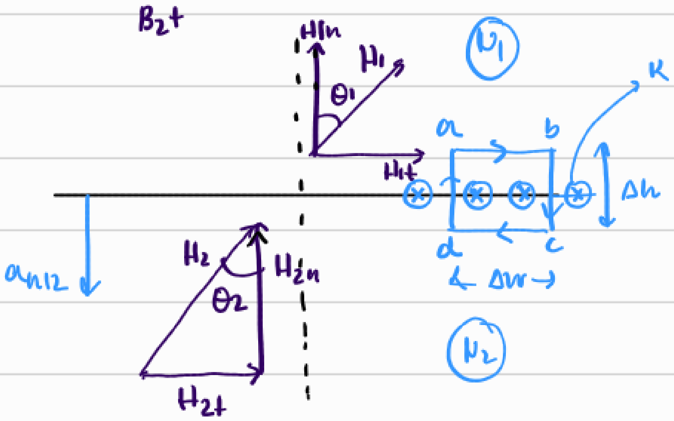
$\oint_L H \cdot dl = I$



$\Rightarrow \oint B \cdot ds = 0$

$\Rightarrow B_{1n} \Delta s - B_{2n} \Delta s = 0$

$\Rightarrow B_{1n} = B_{2n} \Rightarrow \mu_1 H_{1n} = \mu_2 H_{2n}$



$K \Delta w = H_{1t} \Delta w + H_{1n} \cdot \cancel{\frac{\Delta h}{2}} + H_{2n} \cancel{\frac{\Delta h}{2}} - H_{2t} \Delta w - H_{2n} \cdot \cancel{\frac{\Delta h}{2}} - H_{1n} \cancel{\frac{\Delta h}{2}}$

as $\Delta h \to 0$

$K \cancel{\Delta w} = (H_{1t} - H_{2t}) \cancel{\Delta w}$

$H_{1t} - H_{2t} = K$

$K = (H_1 - H_2) \times a_{n12}$ → unit vector normal to the surface directed from med 1 to med 2

If fields make an angle θ_1 and θ_2 with the normal to the surface:

$B_1 \cos\theta_1 = B_{1n} = B_{2n} = B_2 \cos\theta_2$

$\frac{B_1}{\mu_1} \sin\theta_1 = \frac{B_2}{\mu_2} \sin\theta_2 = H_{1t} = H_{2t}$

$$\frac{\tan\theta_1}{\tan\theta_2} = \frac{\mu_1}{\mu_2}$$

TIME VARYING FIELDS:

↳ what happens when electric and magnetic field change with time

→ when fields vary with time:

↳ changing magnetic field creates electric field

↳ changing electric field creates magnetic field

$\frac{\partial B}{\partial t} \rightarrow E \quad ; \quad \frac{\partial E}{\partial t} \rightarrow H$

Faraday's law:

↳ changing magnetic flux induces emf

$$\Psi_B = \int B \cdot ds$$

$$\mathcal{E}\,(\text{Emf}) = -\frac{d\Psi_B}{dt} \rightarrow$$ where $\mathcal{E} \rightarrow$ Emf

$\Psi_B \rightarrow$ magnetic flux

$(-) \rightarrow$ lenz law

magnetic flux → changes → current gets induced

Integral form: $\oint \vec{D} \cdot \vec{ds} = \int \rho v \cdot dv = Q_{enc}$ | $\oint H \cdot dl = \int J \cdot ds = I_{enc}$

Point form: $\nabla \cdot D = \rho v$ | $\nabla \times H = J$

$\nabla \circ K \rightarrow$ Gradient → scalar

$\nabla \times \vec{A} \rightarrow$ Curl

$\nabla \circ \vec{A} \rightarrow$ divergence

] → vector

$\vec{E} \rightarrow V/m$

$\vec{D} \rightarrow C/m^2$

$\vec{H} \rightarrow A/m$

$\vec{B} \rightarrow Wb/m^2$

sources are not changing their magnitude

Maxwell's final Time constant field equations:

↳ (1) $\vec{\nabla} \times \vec{H} = J$ $\qquad \oint H \cdot dl = \int_s J \cdot ds$

(2) $\vec{\nabla} \times \vec{E} = 0$ $\qquad \oint_c E \cdot dl = 0$

(3) $\vec{\nabla} \cdot \vec{D} = \rho v$ $\qquad \oint_s \vec{D} \cdot ds = \int_v \rho v \, dv$

(4) $\vec{\nabla} \cdot \vec{B} = 0$ $\qquad \oint B \cdot ds = 0$

Continuity equation:

→ Consider the time varying +ve charges are continuosly distributed in a region of volume with charge density of ρv

Total charge in the region at a particular time $Q(t)$

net current flowing out of the region

$$I = I_2 - I_1$$

$$I = (-)\frac{d}{dt}(Q(t))$$

charge dec wrt time

$$I = \oint_L J \cdot ds = Q(t) = \int_V \rho v \cdot dv$$

$$\oint_S J \cdot ds = -\frac{d}{dt}\int_V \rho v \cdot dv$$

divergence theorem

$$\int_V \nabla \cdot J \cdot dv = -\int_V \frac{\partial}{\partial t}(\rho v \cdot dv)$$

$$\nabla \cdot J = -\frac{\partial \rho v}{\partial t}$$

Modified Maxwell's equation: [Ampere's law]

$$\nabla \times H = J$$

$J_d = \frac{\partial D}{\partial t}$ → displacement current → changing electric field acts like current

$$\nabla \times H = J + J_d$$

Integral form derivation of Faraday's law:

$$\oint_L E \cdot dl = -\frac{d}{dt}\int B \cdot ds$$

$$\int_S \nabla \times E \cdot ds = -\int \frac{\partial B}{\partial t} \cdot ds$$

$\therefore \nabla \times E = -\frac{\partial B}{\partial t}$ → differential form

---

time varying fields (or) waves are due to accelerated charges (or) time varying currents

Any pulsating current will produce radiation [square wave, rectangular wave]

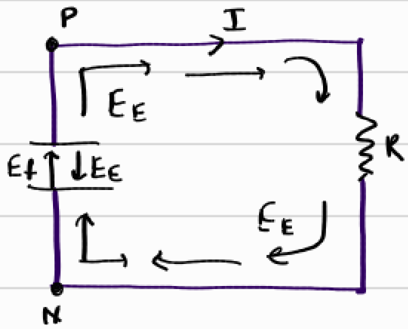
According to Faraday's experiment, a static magnetic field produces no current flow but in a closed circuit, a time varying field produces an induced voltage → electro motive force (emf)) → causes a flow of current

$$V_{emf} = -\frac{d\lambda}{dt} = -N\frac{d\Psi}{dt}$$

acts in a way to oppose the flux producing it

→ $\lambda = N\Psi$ (flux linkage)
where N is the number of turns in the circuit
→ Ψ is the flux through each turn

Lenz's law: the direction of current flow in the circuit is such that the induced magnetic field produced by the induced current opposes change in the original magnetic field



→ $E_E = -\nabla V$
$E_f \to 0$ outside the battery

$$\oint_L E\cdot dl = \oint_L E_f\cdot dl + \cancel{\oint E_E\cdot dl}^{\,0} = \int_N^P E_f\cdot dl$$

↳ conservative

$$V_{emf} = \int_N^P E_f\cdot dl = -\int_N^P E_E\cdot dl = IR$$

→ since E_E and E_f are equal and opposite inside the battery

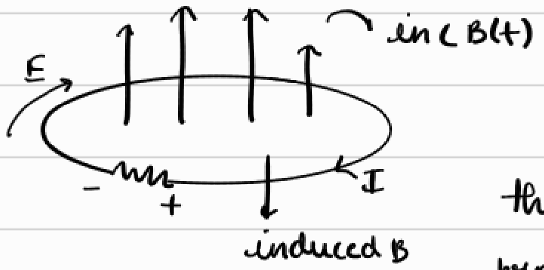
E_f → non conservative

TRANSFORMER AND ELECTROMOTIVE FORCE: ↱ maxwell's curl equation for electrostatic fields because of magnetic field

$$V_{emf} = -\frac{d\Psi}{dt} \quad \text{for } N=1$$

$$V_{emf} = \oint_L E\cdot dl = -\frac{d}{dt}\int_S B\cdot ds$$

(A) Stationary loop in time varying B field (Transformer emf)



$$V_{emf} = \oint_L E \cdot dl = -\int_S \frac{\partial B}{\partial t} \cdot ds$$

the emf produced by a time varying current producing a time varying field in a stationary loop is referred to as transformer emf

Apply stokes theorem, we get

$$\int_S (\nabla \times E) \cdot ds = -\int_S \frac{\partial B}{\partial t} \cdot ds \Rightarrow \nabla \times E = -\frac{\partial B}{\partial t} \rightarrow ① \text{ Maxwell's eq}^n \text{ of time varying fields}$$

(B) Moving loop in a static B field (motional emf)

force on a charge Q moving with uniform velocity u in a magnetic field B

$$F_m = Qu \times B$$

motional electric field : $E_m = \frac{F_m}{Q} = u \times B$

conducting loop moving with uniform velocity

$$V_{emf} = \oint E_m \cdot dl = -\int \frac{\partial B}{\partial t} \cdot ds$$

$$= \oint (u \times B) \cdot dl \longrightarrow \text{motional emf / flux cutting emf}$$

If u and B are perpendicular

$F_m = I l B \longrightarrow I \cdot \cancel{t} \cdot l/\cancel{t} \cdot B = I l B$

$V_{emf} = u l B$

Applying Stokes theorem

$$\oint_S (\nabla \times E_m) \cdot ds = \oint_S \nabla \times (u \times B) \cdot ds$$

$$\nabla \times E_m = \nabla \times (u \times B)$$

(c) Moving loop in time varying field: Both transformer and motional emf are present

$$V_{emf} = \oint_L E \cdot dl = -\int_S \frac{\partial B}{\partial t} \cdot ds + \oint_L (u \times B) \cdot dl$$

Applying stokes theorem:

$$\int_S (\nabla \times E) \cdot ds = -\int_S \frac{\partial B}{\partial t} \cdot ds + \int_S \nabla \times (u \times B) \cdot ds$$

$$\nabla \times E = -\frac{\partial B}{\partial t} + \nabla \times (u \times B)$$

DISPLACEMENT CURRENT

$$\nabla \times H = J$$

$\nabla \cdot (\nabla \times H) = 0 = \nabla \cdot J$ → divergence of curl of any vector field should be 0

$$\nabla \cdot J = -\frac{\partial \rho_v}{\partial t} \neq 0$$

but the continuity of current requires that

$$\nabla \cdot J = -\frac{\partial \rho_v}{\partial t} \neq 0$$

$$\nabla \times H = J + J_d$$

$$\nabla \cdot (\nabla \times H) = 0 = \nabla \cdot J + \nabla \cdot J_d$$

$$\therefore \nabla \cdot J_d = -\nabla \cdot J = \frac{\partial \rho_v}{\partial t} = \frac{\partial}{\partial t}(\nabla \cdot D) = \nabla \cdot \frac{\partial D}{\partial t}$$

$$\therefore J_d = \frac{\partial D}{dt}$$

$\therefore \quad \nabla \times H = J + \frac{\partial D}{\partial t}$ —— (2) Maxwell's equation for a time varying fields

$J_d = \frac{\partial D}{}$ → displacement current density → apparent current due to changing electric field

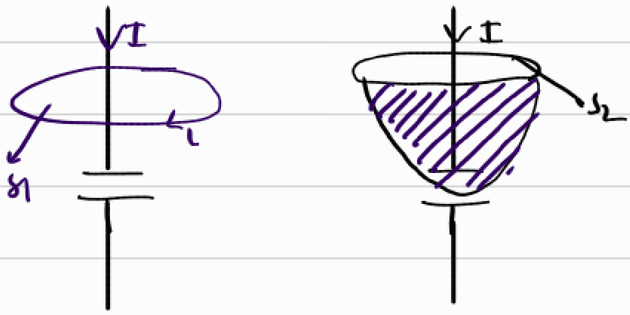
∂t

$J \rightarrow$ conduction current density $\rightarrow$ conduction current is the real current that flows through a conductor $[I = dQ/dt]$

$I_d = \int_{s_1} J_d \cdot ds = \int \frac{\partial D}{\partial t} \cdot ds \rightarrow$ displacement current

$\rightarrow$ as a result of time varying electric field

An eg $\rightarrow$ charging a capacitor:



We know

$\oint H \cdot dl = I_{enc} = \int_{s_1} J \cdot ds = I$

$\oint H \cdot dl = I_{enc} = \int_{s_2} J \cdot ds = 0$

as there is no conduction path

$\therefore$ to resolve this conflict. we include I_d in Ampere's Circuit law

$J_{tot} = J + J_d$

$\therefore \oint H \cdot dl = \int_{s_2} J \cdot ds = \frac{\partial}{\partial t}\int_{s_2} D \cdot ds = \frac{dQ}{dt} = I$ $\rightarrow$ although in $s_1 \rightarrow$ conduction current; $s_2 \rightarrow$ displacement current

MAXWELL'S EQUATION IN THE FINAL FORM:

LORENTZ FORCE EQUATION: $F = Q[E + u \times B]$

CONTINUITY EQUATION: $\nabla \cdot J = -\frac{\partial \rho_v}{\partial t}$

CONSTITUTIVE RELATIONS: $D = \varepsilon E = \varepsilon_0 E + P$

$B = \mu H = \mu_0 [H + M]$ $\rightarrow M = (\mu_r - 1)H$

$J = \sigma E + \rho_v H$

BOUNDARY CONDITIONS: $E_{1t} - E_{2t} = 0$ or $(E_1 - E_2) \times a_n = 0$

$H_{1t} - H_{2t} = K$ or $(H_1 - H_2) \times a_n = K$

$D_{1n} - D_{2n} = \rho_s$ or $(D_1 - D_2) \cdot a_n = \rho_s$

$B_{1n} - B_{2n} = 0$ or $(B_1 - B_2) \cdot a_n = 0$

PERFECT CONDUCTOR: $E = 0$ $H = 0$ $J = 0$ $B_n = 0$ $E_t = 0$

COMPATIBILITY EQUATIONS: $\nabla \cdot B = -\rho_m = 0$

$$\nabla \times E = -\frac{\partial B}{\partial t} + \nabla \times (u \times B)$$

CONSTITUTIVE EQUATIONS: $B = \mu H$

$D = \varepsilon E$

EQUILIBRIUM EQUATIONS: $\nabla \cdot D = \rho_v$

$$\nabla \times H = J + \frac{\partial D}{\partial t}$$

Differential Form	Integral Form	Remarks
$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_v \rho_v \, dv$	Gauss's law
$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$	Nonexistence of isolated magnetic charge*
$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_L \mathbf{E} \cdot d\mathbf{l} = -\frac{\partial}{\partial t} \int_S \mathbf{B} \cdot d\mathbf{S}$	Faraday's law
$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{S}$	Ampère's circuit law

TIME VARYING POTENTIALS:

$\nabla \cdot B = 0$ → holds for time varying fields

$B = \nabla \times A$ → holds for time varying situations

Combining faraday's law:

$$\nabla \times E = -\frac{\partial B}{\partial t} = -\frac{\partial (\nabla \times A)}{\partial t}$$

$$\partial t \qquad \partial t$$

$$\nabla \times \left(E + \frac{\partial A}{\partial t}\right) = 0$$ → if the curl of the vector field is 0, that field can be expressed as the negative gradient of a scalar function

$$E + \frac{\partial A}{\partial t} = -\nabla V$$

(or) $$E = -\nabla V - \frac{\partial A}{\partial t}$$

take divergence = $$\nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot A) = -\frac{\rho v}{\varepsilon}$$

$$\nabla^2 A - \nabla(\nabla \cdot A) = -\mu J + \mu\varepsilon \frac{\partial}{\partial t}\nabla V + \mu\varepsilon \cdot \frac{\partial^2 A}{\partial t^2}$$

$$\nabla \cdot A = -\mu\varepsilon \cdot \frac{\partial V}{dt}$$ → Lorentz condition

$$\nabla^2 V - \frac{\mu\varepsilon\partial V^2}{\partial t^2} = -\frac{\rho v}{\varepsilon}$$

$$\nabla^2 A - \mu\varepsilon\frac{\partial^2 A}{\partial t^2} = -\mu J$$

$$\left.\begin{array}{l} V = \displaystyle\int \frac{[\rho v] \cdot \partial v}{4\pi\varepsilon R} \\ A = \displaystyle\int \frac{\mu [J] \partial v}{4\pi R} \end{array}\right] \rightarrow t' = t - R/u$$

retarded time

$$u = \frac{1}{\sqrt{\mu\varepsilon}}$$

TIME HARMONIC FIELDS:

↳ time harmonic fields → varies periodically (or) sinusodially with time

↳ a phasor is a complex number that contains amplitude and the phase of a sinusodial oscillation

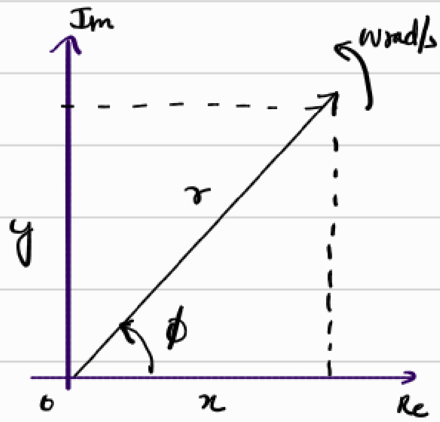
$$z = x + jy = r\angle\phi$$

$j\phi$

$$z = re^{j\phi} = r(\cos\phi + j\sin\phi)$$

$$|r| = |z| = \sqrt{x^2+y^2}$$

$$\phi = \tan^{-1}\left(y/x\right)$$



OPERATIONS: Addition: $z_1 + z_2 = (x_1+x_2) + j(y_1+y_2)$

subtraction: $z_1 - z_2 = (x_1-x_2) + j(y_1-y_2)$

Multiplication: $z_1 * z_2 = (x_1+jy_1)(x_2+jy_2)$

$$= x_1x_2 + jx_1y_2 + jy_1x_2 - y_1y_2$$

$$= (x_1x_2 - y_1y_2) + j(x_1y_2 + x_2y_1)$$

$$z_1 \cdot z_2 = r_1 \cdot r_2 \underline{\angle \phi_1 + \phi_2}$$

division: $\dfrac{z_1}{z_2} = \dfrac{r_1}{r_2} \underline{\angle \phi_1 - \phi_2}$

square root: $\sqrt{z} = \sqrt{r}\, \underline{\angle \phi/2}$

complex conjugate: $z^* = x - jy = r\underline{\angle -\phi} = re^{-j\phi}$

To introduce the time element:

$\phi = wt + \theta$ / $\mapsto$ θ is a fn of time/space/constant

$$re^{j\phi} = re^{jwt+\theta} = re^{j\theta} \cdot e^{jwt}$$

$$Re\{re^{j\phi}\} = r\cos(wt+\theta)$$

$$Im\{re^{j\phi}\} = r\sin(wt+\theta)$$

(eg) $I(t) = I_0 \cos(wt+\theta)$ (sinusodial current)

↳ is the real part of $I_0 e^{j\theta} \cdot e^{jwt}$

complex term $I_0 e^{j\theta}$ is called the phasor current

$I_s = I_0 e^{j\theta} = I_0 \underline{\angle \theta}$ → may represent a scalar (or) vector quantity

$$\therefore I(t) = Re\{I_s \cdot e^{jwt}\}$$

If A is time harmonic:

$$A(x,y,z,t) = Re\{A_s(x,y,z,t) \cdot e^{j\omega t}\}$$

$$A = Re\{A_0 e^{-j\beta x} a_y \cdot e^{j\omega t}\}$$

$$A_s = A_0 e^{-j\beta x} \cdot a_y$$

$$A = Re\{A_s \cdot e^{j\omega t}\}$$

$$\frac{\partial A}{\partial t} = Re\{j\omega A_s e^{j\omega t}\} \Rightarrow \frac{\partial A}{\partial t} \longrightarrow A_s \cdot j\omega$$

similarly: $\int A \cdot dt = \dfrac{A_s}{j\omega}$

$$\nabla \times E(x,y,z,t) = \frac{-\partial B(x,y,z,t)}{dt}$$

let $E(x,y,z,t) = Re\{E_s(x,y,z) \cdot e^{j\omega t}\}$

$B(x,y,z,t) = Re\{B_s(x,y,z) \cdot e^{j\omega t}\}$

$$\nabla \times Re\{E_s e^{j\omega t}\} = \frac{-\partial}{\partial t}(Re\, B_s e^{j\omega t})$$

$$\nabla \times Re\{E_s e^{j\omega t}\} = -B_s \cdot j\omega$$

$$\nabla \times \{Re\{E_s e^{j\omega t}\}\} = -j\omega B_s$$

$$\nabla \times E_s = -j\omega B_s$$

Point Form	Integral Form
$\nabla \cdot \mathbf{D}_s = \rho_{vs}$	$\oint \mathbf{D}_s \cdot d\mathbf{S} = \int \rho_{vs}\, dv$
$\nabla \cdot \mathbf{B}_s = 0$	$\oint \mathbf{B}_s \cdot d\mathbf{S} = 0$
$\nabla \times \mathbf{E}_s = -j\omega \mathbf{B}_s$	$\oint \mathbf{E}_s \cdot d\mathbf{l} = -j\omega \int \mathbf{B}_s \cdot d\mathbf{S}$
$\nabla \times \mathbf{H}_s = \mathbf{J}_s + j\omega \mathbf{D}_s$	$\oint \mathbf{H}_s \cdot d\mathbf{l} = \int (\mathbf{J}_s + j\omega \mathbf{D}_s) \cdot d\mathbf{S}$